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Impact Analysis of Population Aging on Public Education Financial Expenditure in China

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Abstract: The problem of China's aging population is becoming increasingly serious, which brings a variety of social and economic issues. This paper constructed the theoretical framework and analyzed the impact mechanism of the population aging on local public education financial expenditure, and selected provincial panel data from 2010 to 2020 for empirical testing. It is found that population aging has a significant negative impact on local public education financial expenditure, and that with every 1% increase in old-age dependency ratio, local public education financial expenditure will decrease by 0.304%. The increase of urbanization rate has a significant positive impact on local public education financial expenditure, and with every 1% increase in urbanization rate, the local public education financial expenditure will increase by 0.215%. The test of impact mechanism shows that local government fiscal revenue and public expenditure for the elderly has the mediating effect in the impact of population aging on local public education financial expenditure, and the mediating effect of public expenditure for the elderly is stronger. The conclusions of the research can help provide policy implications for solving the declining level of local government public education financial expenditure in the context of an ageing population, and provide theoretical support to improve the rational allocation of fiscal public expenditure.

Keywords: population aging; public education financial expenditure; panel data; fixed-effects model; mediating effect

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1. Introduction

In 2000, China's population aging rate exceeded 7%, officially entering an aging society. By 2021, the population aging rate has grown to 14.2%. With the acceleration of the aging process, the proportion of the labor force in China was also decreasing. From 2010 to 2021, the proportion of the labor force decreased from 74.5% to 62.5%. Changes in the demographic structure will bring about social issues, such as the supply and demand of labor, education, medical care, and old-age care [1].

Moreover, as the population aging rate is on the rise, China's local public education fiscal expenditure has continued to increase in recent years, but which has had no significant improvement on the proportion of the total fiscal expenditure (Figure 1). Due to the different needs and specific distribution of public financial expenditures in different regions, there is also a certain gap in their public education financial expenditures. Considering the current situation of local population aging (Figure 2), the higher the degree of population aging, the lower the proportion of public education financial expenditure.

In order to explore the impact of population aging on public education financial expenditure, scholars at home and abroad have conducted a large number of studies, and the research conclusions can be divided into the following two types.

(1) One argument is that the aging population would hinder financial expenditure on public education. Scholars who hold this view mostly studied from the perspective of

structural changes on population age. For example, Poterba studied the relationship between population age structure and government education expenditure in different regions of the U.S. based on the panel data over the period of 1960–1990, and found that changes in the number of school-age population had little impact on education expenditure, while an increase of the proportion of the elderly would significantly reduce the local education expenditure [2]. By studying the education funding budget system of the Swiss, Grob and Wolter found that the increase in the proportion of the elderly had a significant negative impact on the public education financial expenditure willingness of each state [3]. Life cycle theory suggests that the public service needs of different age groups differ, so there may be intergenerational conflicts in the allocation of public resources. Some scholars have used intergenerational conflict theory to explain the relationship between population aging and public education expenditure [4], verifying the existence of intergenerational conflict in public education expenditure. For example, starting from the relationship between population aging and economic growth, Liu and He set up an OLG model for regression analysis and found that China's public education expenditure was being squeezed out by other public expenditures such as health care and social security with the deepening of population aging [5]. By studying the provincial panel data from the year of 2001 to 2010, Tao and Zhang found that people's support for education expenditure would gradually decline with the trend of population aging, while they were more inclined to social security and pension expenditure [6].

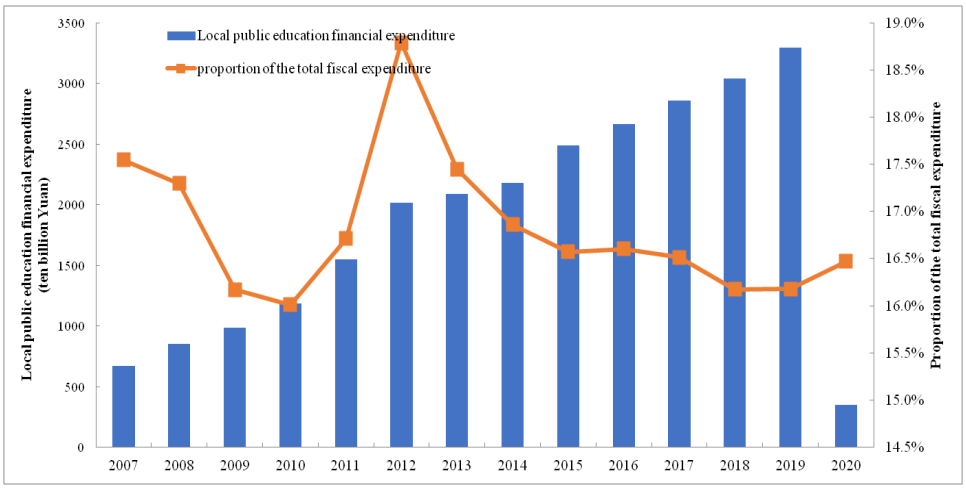

Figure 1. Situation of local public education financial expenditure in China, 2007–2020.

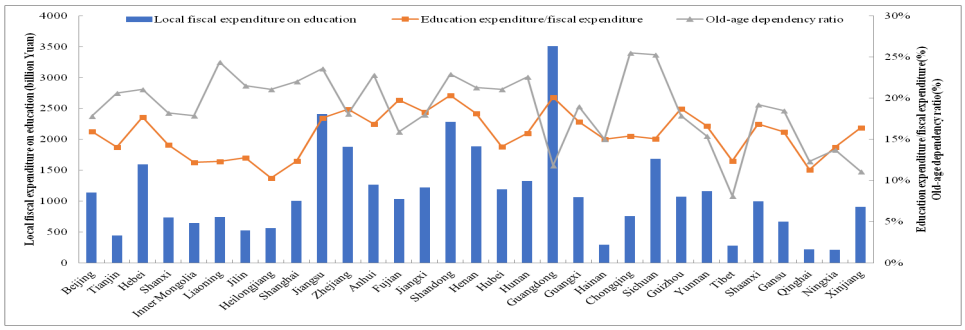

Figure 2. Specific local financial expenditure on education in China in 2020.

In addition to the structural changes of population age, some scholars have conducted relevant studies from the perspective of group interests. The increasing proportion of the elderly will increase the financial pressure on the government and lead to the continued

increase of expenditures on social security and health care. However, those studies only explored from the perspective of theoretical level and did not go into the part of public financial expenditure on education [7]. From the perspective of interest groups, Miller regarded the elderly and the middle-aged as two interest groups, and found that the increase in the proportion of the elderly would reduce the government's education expenditure through regression analysis by the empirical study of the U.S.

(2) The other argument is that population aging would boost public education financial expenditure. By analyzing the data from more than 9000 school districts in the United States, Michael et al. found that the increase in the proportion of native elderly would be beneficial to the local public education financial expenditure [8]. Eric et al. also found that the support of the citizens in the U.S. for boosting educational financial expenditures would increase significantly as their ages increase to 60 and 70 years old [9]. Similar results have also been found in other regions. In Brazil, Arvate and Zoghbi conducted a relevant empirical study based on the panel data of 2054 municipalities, it revealed that the growth of the aging population promoted local education expenditure [10]. Some of the studies have been conducted from the perspective of motivation, and the results show that the elderly believe that public education spending should be increased due to "altruistic" and "mutually beneficial" considerations [11]. Gu also suggested that, due to the deep influence of Confucian culture in China, people are more inclined to reciprocity and mutual benefit than that in the West, and that there is no obvious competition for benefits between different age groups [12].

According to the previous studies, it can be seen that under different institutional and cultural backgrounds, the relationship between population aging and public education financial expenditure was uncertain. Due to differences in research methods and data samples, the direction of the impact of demographic change on public expenditure on education is not consistent, and the relationship between the two needs to be further explored in the cultural context of China. Besides, existing studies have mainly focused on developed countries, the only relevant studies for China have looked at the support for education spending by different age groups from the macro level of overall demographic change and overall national public spending, while specific studies on the impact of population ageing on local public education financial spending lack theoretical analysis and empirical testing. In addition, most of the existing studies have analyzed the unilateral impact of population ageing on public education expenditure without exploring the specific mechanisms of interaction between the variables, and few studies have introduced mediating variables for exploration. The main contributions and innovations of this paper are as follows. Firstly, by constructing a theoretical framework, we analyze the internal impact mechanism of China's population aging on the level of local public education financial expenditure. Secondly, based on the provincial panel data from 2010 to 2020, this paper empirically tests the actual impact of China's aging population on local public education financial expenditure, providing Chinese experience on the issue. Thirdly, in the empirical process, based on the mediating effect model, the mediating role of local government fiscal revenue as well as elderly public expenditure is examined, making the findings of the empirical test more substantial, more convincing and more consistent with the Chinese social context.

2. Theoretical Analysis and Research Hypothesis

Population aging will affect the public education financial expenditure through two basic paths: local government fiscal revenue and local government elderly public expenditure. The influence mechanism of population aging on local public education financial expenditure is shown in Figure 3.

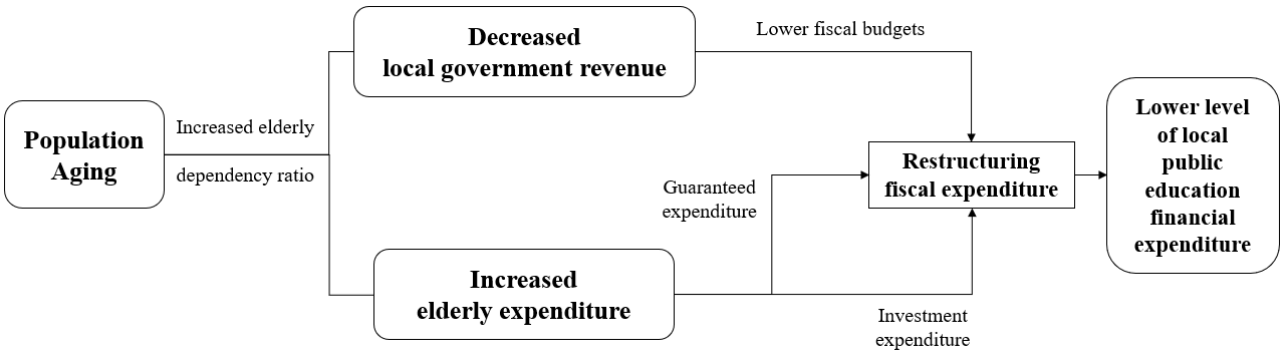

Figure 3. Influence mechanism of population aging on local public education financial expenditure.

Basic Path I: decreased local government revenue. With the increasing proportion of the elderly population, the proportion of the working population is decreasing correspondingly. Such a shift will change the labor productivity, technological innovation, and production and consumption structure of the society, thereby affecting the government's tax revenue and reducing the public fiscal budget [13]. In this case, the government needs to redistribute the share of fiscal expenditure in various fields, which leads to the reduction of the level of local public education financial expenditure.

Basic Path II: increased elderly expenditure. With the ageing of the population, the demand for social security, health care, recreation and other public services for the elderly is increasing. To accommodate the aging society, local governments need to put more financial support on the elderly groups and increase the guaranteed and investment expenditures for them. On the one hand, governments are required to increase expenditures to optimize the health and medical conditions. On the other hand, to meet the growing social needs, governments need to expand investment spending on the elderly industry. For Examples, increasing the production and investment in elderly entertainment products, elderly living products, and elderly health care products, and investing in more aged care facilities. Under a certain scale of expenditure, local governments have to adjust the structure of fiscal expenditure, which will reduce the level of local public education financial expenditure.

Based on the theoretical analysis, the following hypothesis about the impact of population aging on public education financial expenditure can be put forward:

H1. *Population aging will have a negative impact on local public education financial expenditure*;

H2. *Local government fiscal revenue and elderly public expenditure play a mediating role in the process of population aging affecting the level of local public education financial expenditure*.

3. Model Construction, Data Description and Source

3.1. Model Construction

To examine the impact of China's aging population on public education financial expenditure, this paper referred to previous related studies and finally developed the following panel model:

$$lnedu_{i,t} = \beta_{11} lnodr_{i,t} + \beta_{12} lnbudget_{i,t} + \beta_{13} lngdp_{i,t} + \beta_{14} lnurb_{i,t} + \beta_{15} lnyoung_{i.t} + \alpha_{i1} + \eta_{t1} + \varepsilon_{i,t1} \quad (1)$$

Among them, i represents each province (region) in China and t represents different periods (years). In order to reduce the absolute differences between the data and to avoid inaccuracies caused by the influence of individual extreme values, all variables are logarithmically treated, which also helps to reduce the covariance and heteroskedasticity of the model. $lnedu_{i,t}$ represents the level of public education financial expenditure in year t for each province and region i, $lnodr_{i,t}$ represents the old-age dependency ratio in year t for each province and region i, $lnbudget_{i,t}$, $lngdp_{i,t}$, $lnurb_{i,t}$ and $lnyoung_{i.t}$ represent the budget revenue, per capita GDP, urbanization rate and youth population ratio in year t for

each province and region i in China, respectively. α_{i1} and η_{t1} are random variables, which denote fixed effects that do not change over time and do not change by region, respectively, $\varepsilon_{i,t1}$ is a random variation term.

The theoretical analysis showed that population aging would have an impact on public financial education expenditure through the mediating effect of local government fiscal revenue as well as elderly public expenditure. Therefore, further mediating effect models are developed, where models (1), (2), and (3) are used to test the mediating effect of local government fiscal revenue, and models (1), (4), and (5) are used to test the mediating effect of public expenditure on the elderly.

$$lnfrd_{i,t} = \beta_{21}lnodr_{i,t} + \beta_{22}lnbudget_{i,t} + \beta_{23}lngdp_{i,t} + \beta_{24}lnurb_{i,t} + \beta_{25}lnyoung_{i,t} + \alpha_{i2} + \eta_{t2} + \varepsilon_{i,t2} \quad (2)$$

$$lnedu_{i,t} = \beta_{31}lnodr_{i,t} + \beta_{32}ln\ frd_{i,t} + \beta_{33}lnbudget_{i,t} + \beta_{34}lngdp_{i,t} + \beta_{35}lnurb_{i,t} + \beta_{36}lnyoung_{i.t} + \alpha_{i3} + \eta_{t3} + \varepsilon_{i,t3} \quad (3)$$

$$lnhealth_{i,t} = \beta_{41}lnodr_{i,t} + \beta_{42}lnbudget_{i,t} + \beta_{43}lngdp_{i,t} + \beta_{44}lnurb_{i,t} + \beta_{45}lnyoung_{i,t} + \alpha_{i4} + \eta_{t4} + \varepsilon_{i,t4} \quad (4)$$

$$lnedu_{i,t} = \beta_{51}lnodr_{i,t} + \beta_{52}ln\ health_{i,t} + \beta_{53}lnbudget_{i,t} + \beta_{54}lngdp_{i,t} + \beta_{55}lnurb_{i,t} + \beta_{56}lnyoung_{i.t} + \alpha_{i5} + \eta_{t5} + \varepsilon_{i,t5} \quad (5)$$

$lnfrd_{i,t}$ represents the level of fiscal revenue in year t for each province and region i, $lnhealth_{i,t}$ represents the public expenditure on the elderly in year t for each province and region i, and the other variables are as described in the previous section.

3.2. Variable Description

After fully considering the availability of data, this paper selected "public education financial expenditure" to measure local public education financial expenditure in China, based on the practice of scholars such as Guifang Shi [14] and Hao Yu [15], and used this indicator as the explanatory variable. Since the indicators of population aging are still relatively single, mainly the old-age dependency ratio and the elderly population ratio [16–18], this paper selected the "old-age dependency ratio" as the core explanatory variable and used the "elderly population ratio" as the explanatory variable for the "old-age dependency ratio" in the model. The final robustness test was conducted by replacing the "old-age dependency ratio" with the "old-age population ratio" [19]. The "local government fiscal revenue" and " elderly public expenditure" were set as mediating variables. Besides, referring to some existing studies, this paper set "general budget revenue", "per capita GDP", "urbanization rate" and "youth population ratio" as control variables.

3.2.1. Explained Variable

Public education financial expenditure (*lnedu*). This variable is expressed as the ratio of total public education fiscal expenditures to total local fiscal expenditures in each province, city, and autonomous region in China. The main reason is the uneven economic development among Chinese regions, the significant differences in the number of educated population and the public education resources among them. If the model is established based on the original amount of local public education financial expenditure, it will lead to deviations in the data analysis, resulting in distorted analysis results. Using this ratio as an explanatory variable can eliminate the impact of fiscal disparities across regions and obtain a more reasonable mathematical model.

3.2.2. Explanatory Variable

Old-age dependency ratio (*lnodr*), i.e., the proportion of the elderly population to the labor force population. The larger the value of this variable, the more serious the phenomenon of population aging in the region, the higher the number of dependents borne by the labor force population, and also the higher pressure on the government. It is one of the important indicators to measure the aging degree of China's population.

3.2.3. Mediating Variables

(1) Local government fiscal revenue (*lnfrd*)

When conducting the budget for public education financial expenditure, local governments need to decide based on the actual fiscal situation. Since the uneven economic development among regions in China, this variable, expressed as the ratio of total local fiscal revenue to total national fiscal revenue, can be a good measure of regional fiscal revenue.

(2) Elderly public expenditure (*lnhealth*)

There are many components of public expenditure on the elderly, among which the local government expenditure on health care can reflect it best, therefore, this variable is expressed as the total local public finance health expenditure.

3.2.4. Control Variables

To have better explanatory power of the model while controlling for other factors that may have an impact on public education financial expenditures in each region, the following control variables are introduced in this paper.

(1) Budget revenue (*lnbudget*)

Local public education financial expenditure is closely related to budget revenue [20], and since the fiscal revenue of different provinces varies, their level of public education expenditure also varies.

For example, for the economically developed provinces, the level of public education fiscal expenditure is higher.

(2) GDP per capita (*lngdp*)

GDP per capita is calculated by dividing the region's GDP by its total population, which reflects the current living standard of people. In general, the higher the GDP per capita, the better the economic status of the residents, and the more they are able to meet their needs for retirement, health care, and education, thus reducing the financial burden of the government for public service provision to a certain extent.

(3) Urbanization rate (*lnurb*)

The urbanization rate is the main indicator of China's urbanization process, i.e., the proportion of urban population in the total population. As China's urbanization gradually advances, urban population will further cluster, which will also require cities to provide a higher level of education supply, further affecting local public education financial expenditures in China.

(4) Youth population ratio (*ln young*)

The percentage of the youth population, to some extent, affects the district's public education financial expenditure. The higher the value of this variable, the more public education financial expenditure that is needed in the region. In addition, this variable can also reflect the future trend of population aging. Therefore, the youth population ratio is included as a control variable.

Table 1 shows the specific description of the explained, explanatory, mediating, and control variables.

3.3. Data Sources and Descriptive Statistics

Considering the availability and comparability of data, this paper finally selects the panel data of 31 provinces, municipalities, and autonomous regions in China from 2010 to 2020, which are mainly obtained from the "*China Statistical Yearbook*", the "*China Education Expenditure Statistical Yearbook*" and the relevant statistical yearbooks of each province or city.

Table 1. Specific description of the variables.

Variable Type	Name	Symbol	Calculation
Explained variable	Public education financial expenditure	*lnedu*	Public education financial expenditure in year *t* for region *i*/Total financial expendi-ture of region *i* in year *t*
Explanatory variable	Old-age dependency ratio	*lnodr*	Number of people over 65 years old in year *t* for region *i*/Number of people aged 15–64 of region *i* in year *t*
Mediating variables	Local government fiscal revenue	*lnfrd*	Fiscal revenue of region *i* in year *t*/National fiscal revenue in year *t*
	Elderly public expenditure	*lnhealth*	Total financial health care expenditure in year *t* for region *i*
Control variables	Budget revenue	*lnbudget*	Total budget revenue of region *i* in year *t*
	GDP per capita	*lngdp*	GDP of region *i* in year *t*/Total population of region *i* in year *t*
	Urbanization rate	*lnurb*	Number of urban population in year *t* for region *i*/Total population of region *i* in year *t*
	Youth population ratio	*lnyoung*	Number of people aged 0–14 in year *t* for region *i*/Total population of region *i* in year *t*

Table 2 shows the descriptive statistical analysis of these data:

Table 2. Descriptive statistics of the original variables.

Variable	Obs	Mean	Std. Dev.	Min	Max
edu	341	16.177	2.675	9.895	22.217
odr	341	14.135	3.75	6.71	25.5
frd	341	1.708	1.367	0.044	7.065
health	341	371.064	262.203	32.04	1772.99
budget	341	2521.889	2196.626	36.65	12,923.85
gdp	341	53,685.03	27,222.291	13,119	164,889.47
urb	341	57.406	13.422	22.67	89.6
young	341	16.598	4.097	8.114	28.737

As shown in Table 2, the minimum value of the explanatory variable "public education financial expenditure" is 9.895% and the maximum value is 22.217%, which indicates that the financial expenditure on education is unbalanced across regions in China, with large regional differences. The mean value of the explanatory variable "old-age dependency ratio" has reached 14.135%, the minimum value is 6.71%, and the maximum value is

25.2%, indicating that the aging trend of China's population is increasing, which also shows obvious regional differences. The control variables: "budget revenue", "GDP per capita", "urbanization rate", and "youth population ratio" all have significant regional imbalances. To eliminate heteroskedasticity, each variable in the equation was logged, and the descriptive statistics of the variables after logging are shown in Table 3.

Table 3. Descriptive statistics of variables.

Variable	Obs	Mean	Std. Dev.	Min	Max
lnedu	341	2.769	0.173	2.292	3.101
lnodr	341	2.614	0.264	1.904	3.239
lnfrd	341	0.188	0.939	−3.121	1.955
lnhealth	341	5.663	0.761	3.467	7.48
lnbudget	341	7.45	0.981	3.601	9.467
lngdp	341	10.78	0.465	9.482	12.013
lnurb	341	4.022	0.242	3.121	4.495
lnyoung	341	2.776	0.264	2.094	3.358

4. Empirical Analysis

4.1. Correlation Analysis

Correlation between two variables is a prerequisite for regression analysis [21], therefore, the selected variables were first analyzed for correlation, as shown in Table 4:

Table 4. Correlation analysis of variables.

	lnedu	*lnodr*	*lnfrd*	*lnhealth*	*lnbudget*	*lngdp*	*lnurb*	*lnyoung*
lnedu	1.000							
lnodr	0.113 **	1.000						
lnfrd	0.516 ***	0.478 ***	1.000					
lnhealth	0.462 ***	0.657 ***	0.747 ***	1.000				
lnbudget	0.468 ***	0.578 ***	0.966 ***	0.853 ***	1.000			
lngdp	−0.045	0.364 ***	0.521 ***	0.421 ***	0.629 ***	1.000		
lnurb	−0.033	0.388 ***	0.579 ***	0.349 ***	0.638 ***	0.861 ***	1.000	
lnyoung	0.221 ***	−0.266 ***	−0.467 ***	−0.107 **	−0.441 ***	−0.586 ***	−0.738 ***	1.000

** $p < 0.05$, *** $p < 0.01$.

Through correlation analysis, it can be seen that the correlation coefficient between "old-age dependency ratio" and "public education financial expenditure" is 0.113, which is significant at the level of 0.05. The correlation is positive but not strong, and there is no regression relationship between the two variables. Among other variables, "local government fiscal revenue", "elderly public expenditure", "budget revenue" and "youth population ratio" are positively correlated with explanatory variables, and are significant at the 0.01 level.

4.2. Base Model Test

In the processing of the panel data, it is necessary to determine which model should be adopted by conducting the Hausman test. The results are shown in Table 5.

Table 5. Hausman test.

	Coef.
Chi-square test value	39.66
p-value	0

The Hausman test shows a *p*-value of 0. Therefore, rejecting the original hypothesis and using the fixed effect model.

4.3. Regression Results Analysis

4.3.1. Baseline Panel Regression

Based on the model (1), regressions were conducted on public education financial expenditures, and the results are shown in Table 6.

Table 6. Regression results.

lnedu	**Coef.**	**St. Err.**	**t-Value**	***p*-Value**	**[95% Conf**	**Interval]**	**Sig**
lnodr	−0.304	0.034	−8.97	0	−0.371	−0.237	***
lnodr	0.06	0.048	1.25	0.214	−0.035	0.154	
lnbudget	−0.033	0.033	−1.01	0.314	−0.097	0.031	
lnurb	0.215	0.099	2.17	0.031	0.02	0.409	**
lnyoung	0.091	0.081	1.12	0.264	−0.069	0.25	
Constant	2.051	0.339	6.06	0	1.385	2.717	***
Mean dependent var	2.769		SD dependent var		0.173		
R-squared	0.827		Number of obs		341		
F-test	17.905		Prob > F		0.000		
Akaike crit. (AIC)	−883.717		Bayesian crit. (BIC)		−860.726		

*** $p < 0.01$, ** $p < 0.05$.

The coefficient of the old-age dependency ratio is −0.304 and significant at the 1% level, which means that an increase in the old-age dependency ratio has a significant negative effect on local fiscal spending on public education. This indicates that the deepening of population aging will reduce local fiscal spending on public education, and with each 1% increase in the old-age dependency ratio, public education financial expenditure will decrease by 0.304%. This regression result also verifies the research hypothesis 1: Population aging will have a negative impact on local public education financial expenditure.

In addition, among the results of the control variables, the coefficient of the urbanization rate is 0.215 and is significant at the 5% level. This indicates that an increase in the urbanization rate will be beneficial to the local government's fiscal spending on public education, i.e., an increase in the urbanization rate has a significant positive effect on public education financial expenditure. With each 1% increase in the urbanization rate, the local public education financial expenditure will increase by 0.215%, which further indicates that the agglomeration of the urban population can effectively promote the increase of local fiscal expenditure on public education. The possible reason for this is that with the large influx of rural population into cities, the government needs to formulate and improve the corresponding supporting policies, which include solving the education problems of children and equipping them with sufficient education resources, thus requiring a continuous increase in public education fiscal spending. In addition, as the urbanization rate increases, the quality of public services faces higher demands.

4.3.2. Impact Mechanism Test

Theoretical analysis suggests that an increase in the old-age dependency ratio will lead to a decrease in local government fiscal revenue and an increase in public spending on older groups, which will reduce the level of public education financial expenditure. To test this impact mechanism, this paper conducts a test, referring to the method of Zhonglin Wen et al. [22] to test the mediating effect.

The test for the mediating variable "local government fiscal revenue" is as follows: (1) test the effect of population aging on local public education expenditure; (2) test whether population aging affects the mediating variable "local government fiscal revenue"; (3) test the effect of population aging and fiscal revenue on local public education financial expenditure at the same time. The test idea of the mediating variable "elderly public expenditure" refers to the above steps.

To test the mediating effect of local government fiscal revenue, models (1), (2), and (3) were regressed respectively, and Table 7 shows the results. The coefficient of the middle-

aged dependency ratio in the model (2) is −0.362, and is significant at the 1% level, i.e., the effect of population aging on public education financial expenditure is significantly negative. This indicates that the deepening of population aging will inhibit the increase of fiscal revenue. In model (3), when both "old-age dependency ratio" and "local government fiscal revenue" are included, population aging still has a significant negative impact on local public education financial expenditure.

Table 7. Regression results.

	(1) ***lnedu***	**(2)** ***lnfrd***	**(3)** ***lnedu***
lnodr	−0.304 ***	−0.362 ***	−0.262 ***
	(0.034)	(0.038)	(0.038)
lngdp	0.060	−0.202 ***	0.083 *
	(0.048)	(0.053)	(0.049)
lnbudget	−0.033	0.558 ***	−0.098 **
	(0.033)	(0.036)	(0.043)
lnurb	0.215 **	−0.216 *	0.240 **
	(0.099)	(0.110)	(0.099)
lnyoung	0.091	0.293 ***	0.057
	(0.081)	(0.090)	(0.082)
lnfrd			0.116 **
			(0.051)
_cons	2.051 ***	−0.791 **	2.143 ***
	(0.339)	(0.377)	(0.339)
N	341	341	341
adj. R^2	0.638	0.603	0.850

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

To test the mediating effect of elderly public spending, models (1), (4), and (5) were regressed respectively, and the results are shown in Table 8. The effect of population aging on public education financial expenditure is significantly negative, and the coefficient of the old-age dependency ratio in the model (4) is 0.457, significant at the level of 1%, which means that an increase in the old-age dependency ratio has a significant positive effect on local public expenditure on health care. This indicates that the deepening of population aging will force the government to boost public spending on the elderly. In model (5), when both "old-age dependency ratio" and "elderly public expenditure" are included, population aging still has a significant negative effect on local public education financial expenditure.

The regression results of Tables 7 and 8 show that local government fiscal revenue and elderly public expenditure are the mediating variables, which affect the local public education fiscal financial expenditure, and the partial mediating effect holds. These also validate the research hypothesis 2.

According to the results of the intermediary effect test and the classification criteria, the proportion of the intermediary effect can be calculated (Table 9). Population aging will affect the level of local public education financial expenditure by affecting fiscal revenue and public expenditure for the elderly, and the proportion of the mediation effect to the total effect is 0.1381 and 0.1563, respectively.

Table 8. Regression results.

	(1) *lnedu*	(4) *lnhealth*	(5) *lnedu*
lnodr	−0.304 ***	0.546 ***	−0.257 ***
	(0.034)	(0.054)	(0.039)
lngdp	0.060	0.538 ***	0.106 **
	(0.048)	(0.077)	(0.051)
lnbudget	−0.033	0.500 ***	0.010
	(0.033)	(0.052)	(0.037)
lnurb	0.215 **	0.729 ***	0.278 ***
	(0.099)	(0.158)	(0.102)
lnyoung	0.091	−0.170	0.076
	(0.081)	(0.129)	(0.081)
lnhealth			−0.087 **
			(0.036)
_cons	2.051 ***	−7.747 ***	1.380 ***
	(0.339)	(0.541)	(0.434)
N	341	341	341
adj. R^2	0.638	0.932	0.652

Standard errors in parentheses. ** $p < 0.05$, *** $p < 0.01$.

Table 9. The proportion of mediation effect.

Explained Variable	Explanatory Variable	Mediating Variable	Test Result	Mediating Effect Proportion
Public education financial expenditure	Old-age dependency ratio	Local government fiscal revenue	Existing	0.1381
		Elderly public expenditure	Existing	0.1563

4.4. Robustness Test

To test the robustness of the panel data and further verify the impact of population aging on public education financial expenditure, a regression analysis was conducted by replacing the core explanatory variable in the model. The specific replacement method is to replace the old-age dependency ratio (*lnold*) with the old-age population ratio (*lnold*). In this case, the old-age population ratio (*lnold*) is expressed as the ratio of the population over the age of 65 in each province i in year t to the total population in province i in year t. The regression results are shown in Table 10, and it can be seen that the results are not significantly different from the previous ones, so the robustness of the panel regression model is good.

Table 10. Robustness test results.

	(1) *lnedu*	(2) *lnfrd*	(3) *lnedu*	(4) *lnhealth*	(5) *lnedu*
lnold	−0.342 ***	−0.486 ***	−0.290 ***	0.670 ***	−0.280 ***
	(0.041)	(0.043)	(0.048)	(0.064)	(0.047)
lngdp	0.056	−0.185 ***	0.076	0.529 ***	0.105 **
	(0.049)	(0.051)	(0.049)	(0.076)	(0.052)
lnbudget	−0.008	0.587 ***	−0.071	0.455 ***	0.035
	(0.033)	(0.035)	(0.046)	(0.051)	(0.037)
lnurb	0.163	−0.236 **	0.188 *	0.793 ***	0.236 **
	(0.099)	(0.104)	(0.100)	(0.155)	(0.103)

Table 10. *Cont.*

	(1) *lnedu*	(2) *lnfrd*	(3) *lnedu*	(4) *lnhealth*	(5) *lnedu*
lnyoung	0.049	0.240 ***	0.023	−0.092	0.040
	(0.082)	(0.086)	(0.083)	(0.128)	(0.081)
lnfrd			0.107 **		
			(0.054)		
lnhealth					−0.093 **
					(0.036)
_cons	2.217 ***	−0.789 **	2.302 ***	−7.905 ***	1.483 ***
	(0.339)	(0.356)	(0.340)	(0.529)	(0.442)
N	341	341	341	341	341
adj. R^2	0.714	0.636	0.623	0.934	0.730

Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5. Conclusions and Policy Recommendations

5.1. Main Research Conclusions

Population aging is the inevitable result of demographic transition and socio-economic development. Based on the reality of China's deepening population aging, this paper studies the impact of population aging on local public education financial expenditure with the provincial panel data from 2010 to 2020, and the main conclusions are as follows. First, the distribution of educational resources is uneven, local financial strength varies widely, and there are disparities in public education fiscal expenditures among different regions. Population aging has a significant negative effect on local public education financial expenditure, with each 1% increase in the old-age dependency ratio, local public education financial expenditure will decrease by 0.304%. Second, the increase of urbanization rate has a significant positive effect on local public education fiscal expenditure, and the concentration of the urban population can effectively promote the increase of local public education fiscal expenditure in China. Third, local government fiscal revenue and elderly public expenditure play a partial mediating role in this process, and the mediating role of elderly public expenditure is stronger. On the one hand, the continued rise in population ageing will depress local government revenue levels, and thus local public education financial expenditure; on the other hand, deepening ageing will boost local public spending on the elderly, which in turn will have a negative impact on public education financial expenditure.

5.2. Policy Recommendations

At present, with the decrease of fertility rate and the increase of life expectancy in China, population aging has become an inevitable trend of social development and will continue to deepen in the future [23]. Therefore, we should face up to and actively deal with the impact of population aging on local governments' public education financial expenditure.

(1) Long-term perspective. China's population age structure is difficult to change in a short period [24], and the impact of population aging on local public education financial expenditure will be long-standing. First, the birth rate raise. All of these problems caused by population aging must be solved from the perspective of population age structure, actively adjusting population age structure and improving fertility level. For example, in terms of maternity policy, the maternity leave and maternity insurance systems should be improved to protect the legitimate rights and interests of women in employment [24]; in terms of child policy, the supply of quality education resources should be promoted to reduce family education expenses, thereby increasing the willingness of the population of childbearing age to have children. Second, accelerate urbanization. Empirical analysis shows that the increase of urbanization rate has a positive effect on local public education financial expenditure. Therefore, it is necessary to push forward the reform of the household registration system in an orderly manner, relax the policy of urban household registration to attract

more talents [25]; establish and improve the basic public service system to protect the basic rights and interests of residents; and constantly improve the infrastructure of housing, transportation, health, and telecommunications in towns and cities to raise the level of urban development.

(2) Short-term perspective. First, increase the level of public education financial expenditure. The central government should increase the proportion of public education fiscal expenditure in GDP [26], and maintain a stable and continuous growth of it. Local governments should also further increase public education fiscal spending and raise its share of local GDP to avoid the compression of education fiscal investment due to aging. Second, central government transfer payments. Taking into account the actual situation of each place, such as regional economic development, current situation of educational resources and population development trends, etc., strengthen the expenditure on areas with less developed and poor educational resources. Third, reconstruct the incentive mechanism of local spending. Under the current fiscal system, local governments have more autonomy [27]. The central government should enhance the supervision of local public education fiscal expenditures and gradually guide local governments to shift from pursuing economic growth to pursuing the provision of public service. For example, set more diverse and comprehensive assessment indicators for local officials [28], release positive signals in relevant policy statements to avoid local governments from blindly pursuing local economic development and neglecting local public education fiscal investment.

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